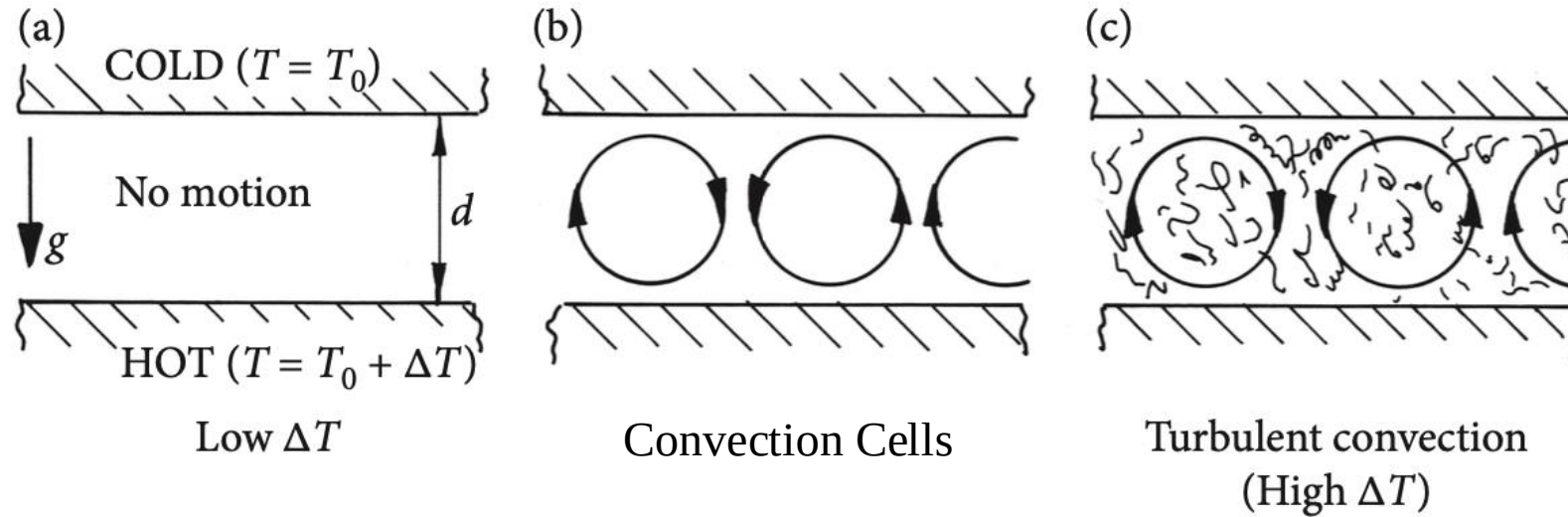


Transition to turbulence



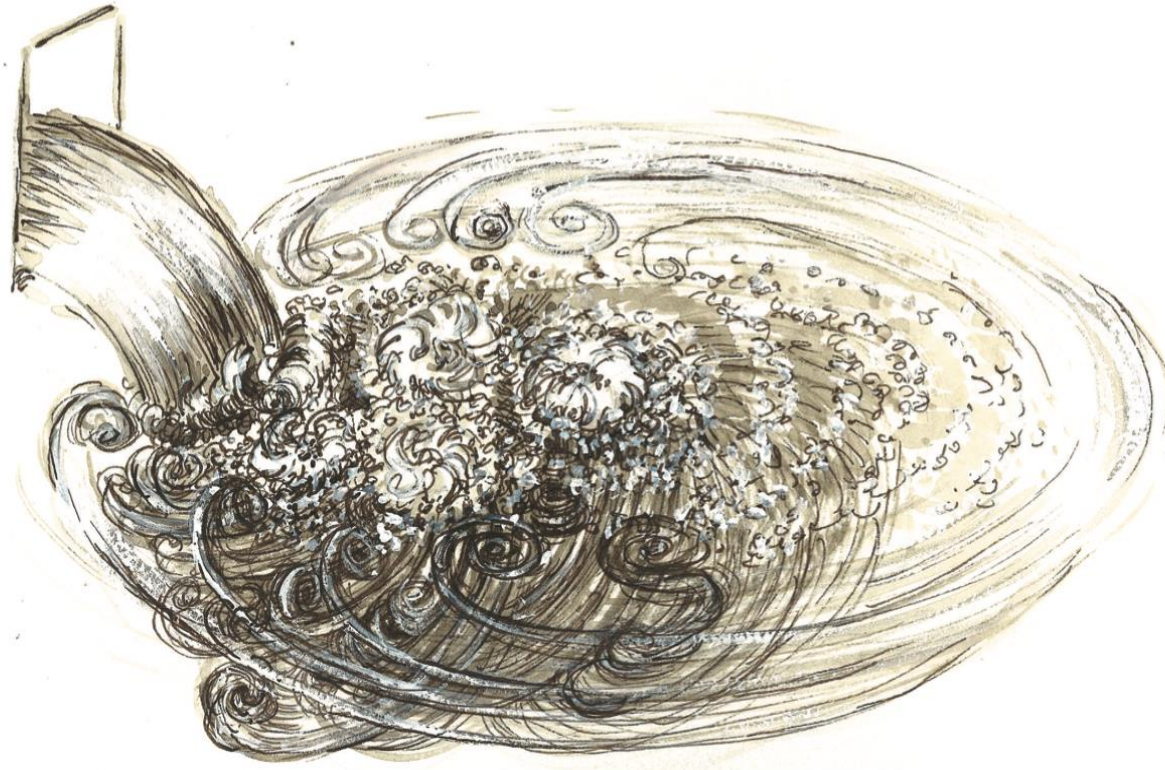
Onset of convection when heating a plate from below.

When the temperature gradient is small, no motion happens.

At a particular value of ΔT , convective cells form.

For very high values of ΔT , the convective cells themselves turn turbulent.

History: “Turbolenza”



“... the smallest eddies are almost numberless, and large things are rotated only by large eddies and not by small ones, and small things are turned by small eddies and large.”

Transition to turbulence

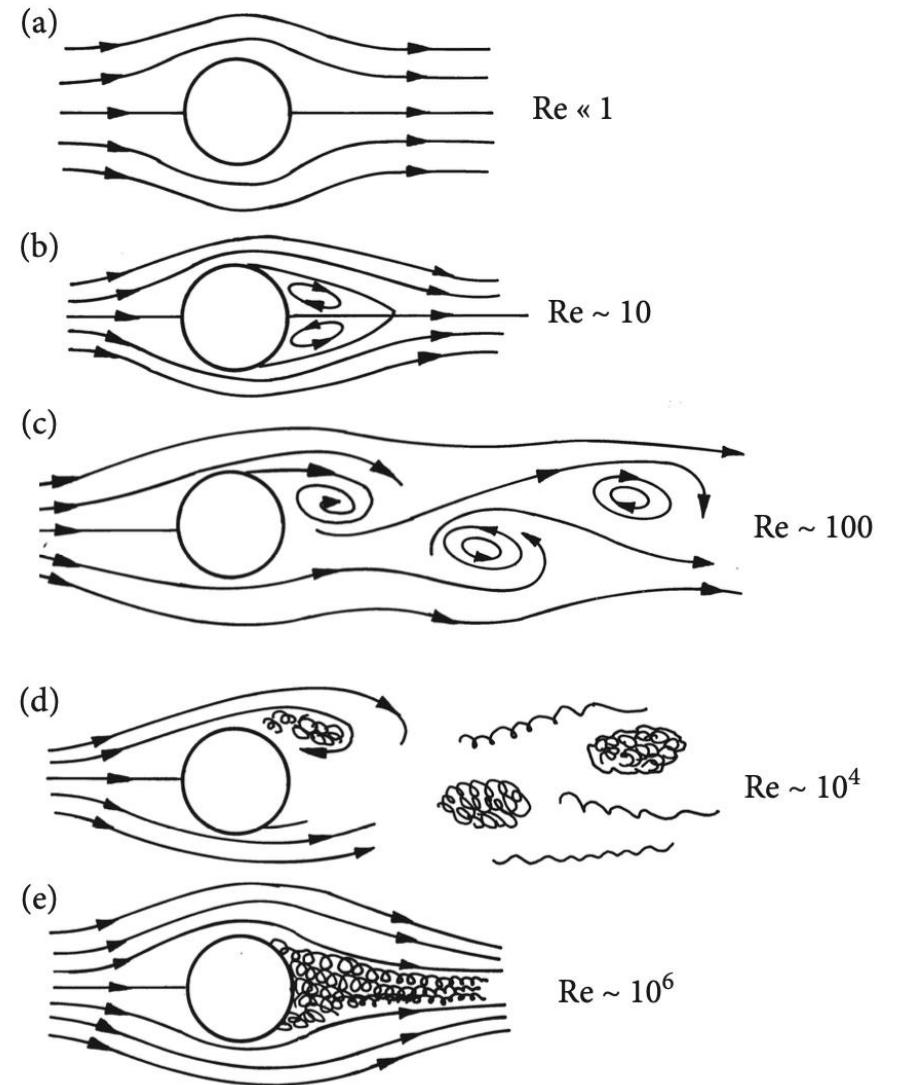
Turbulence is the norm, not the exception.

In dimensionless form, the NS equation is

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \frac{1}{\text{Re}} \nabla^2 \mathbf{u} + \mathbf{f}$$

The Reynolds number is the only parameter by which two systems differ.

It is only by the action of viscosity that turbulent flow is suppressed.

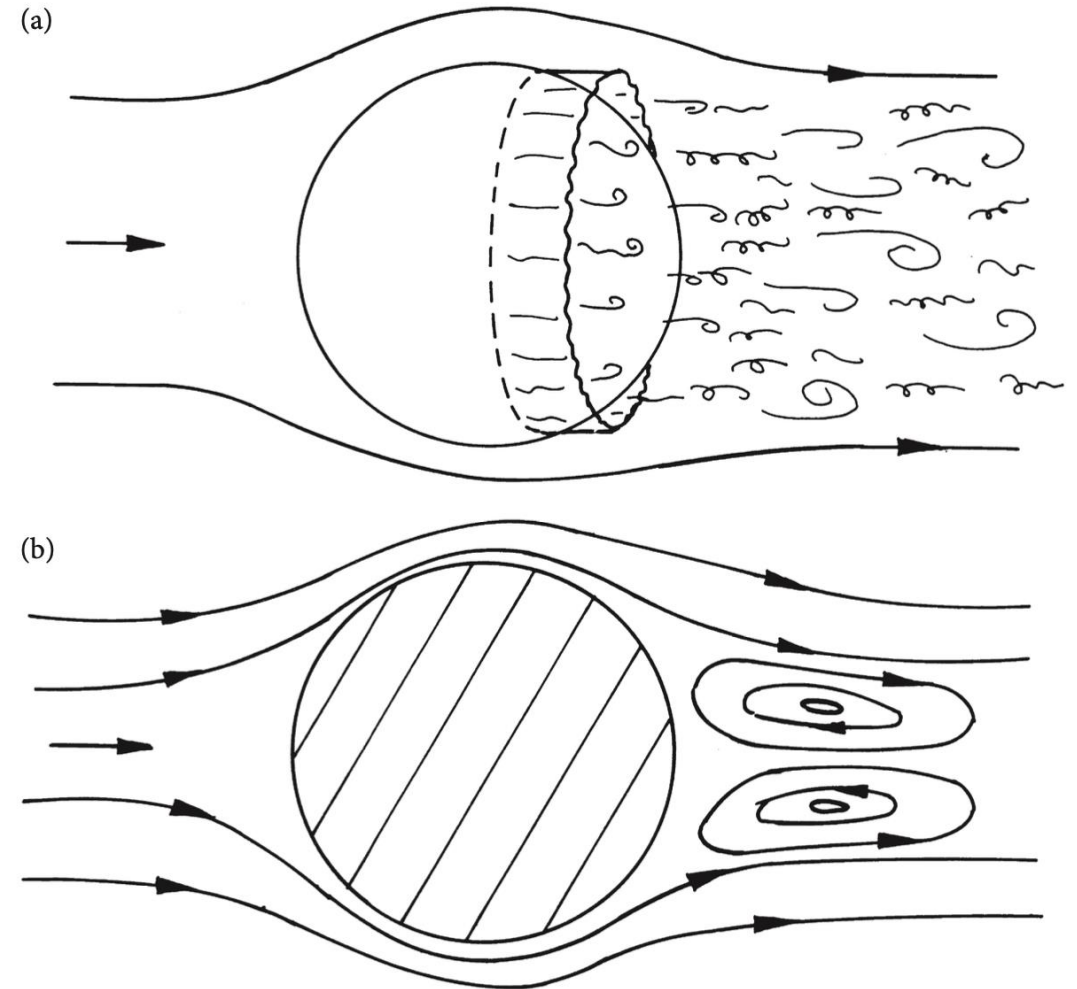
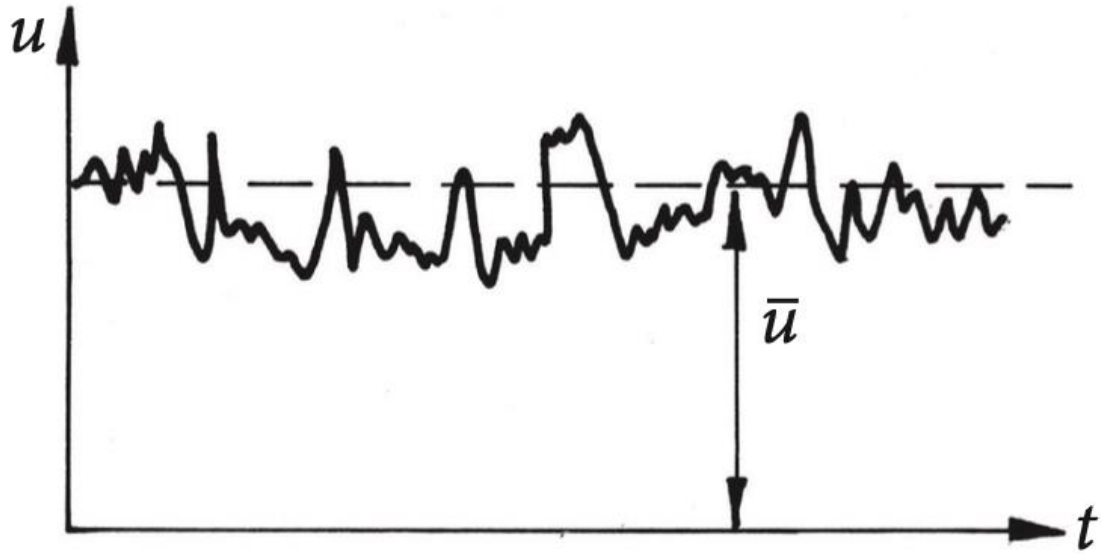


Transition to turbulence

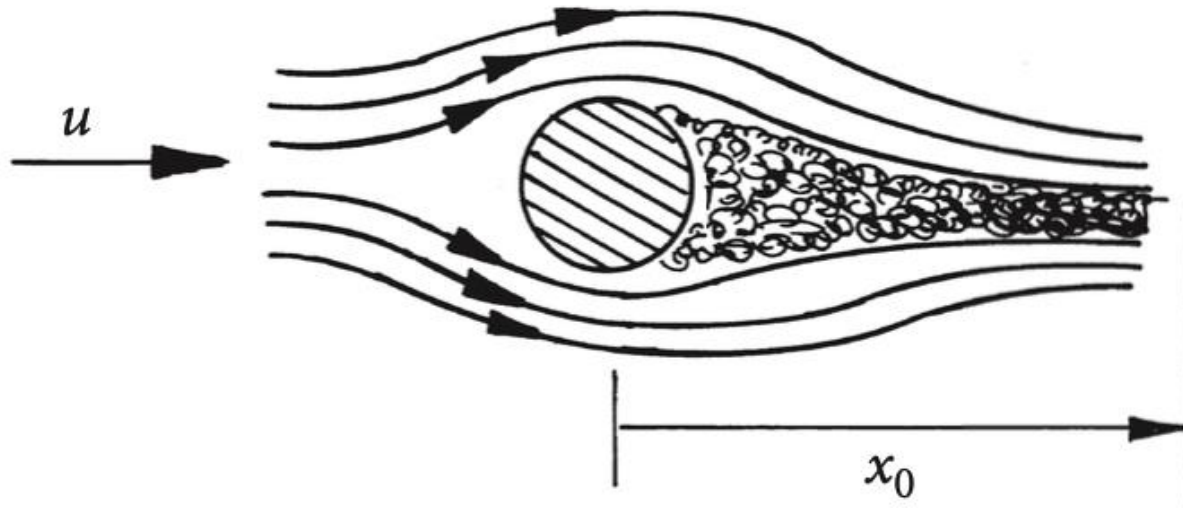
As cigarette smoke rises, it accelerates.
Eventually the Reynolds number
becomes large enough.



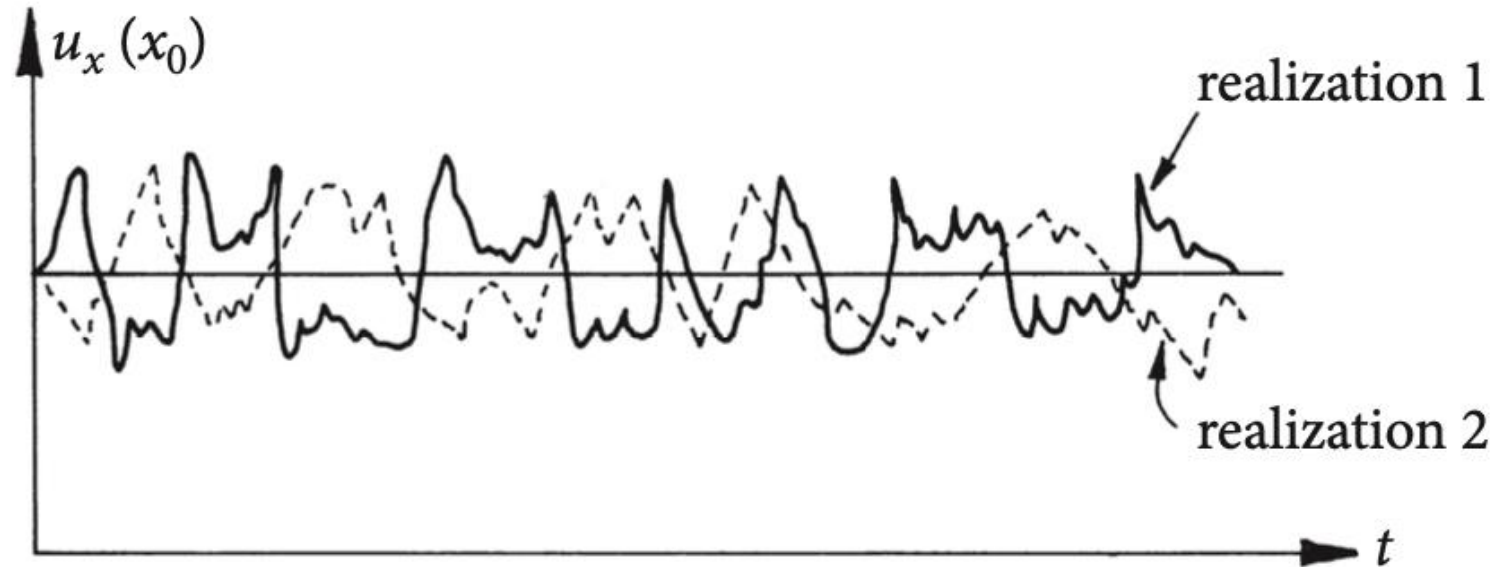
Mean flow and perturbation



Mean flow and perturbation



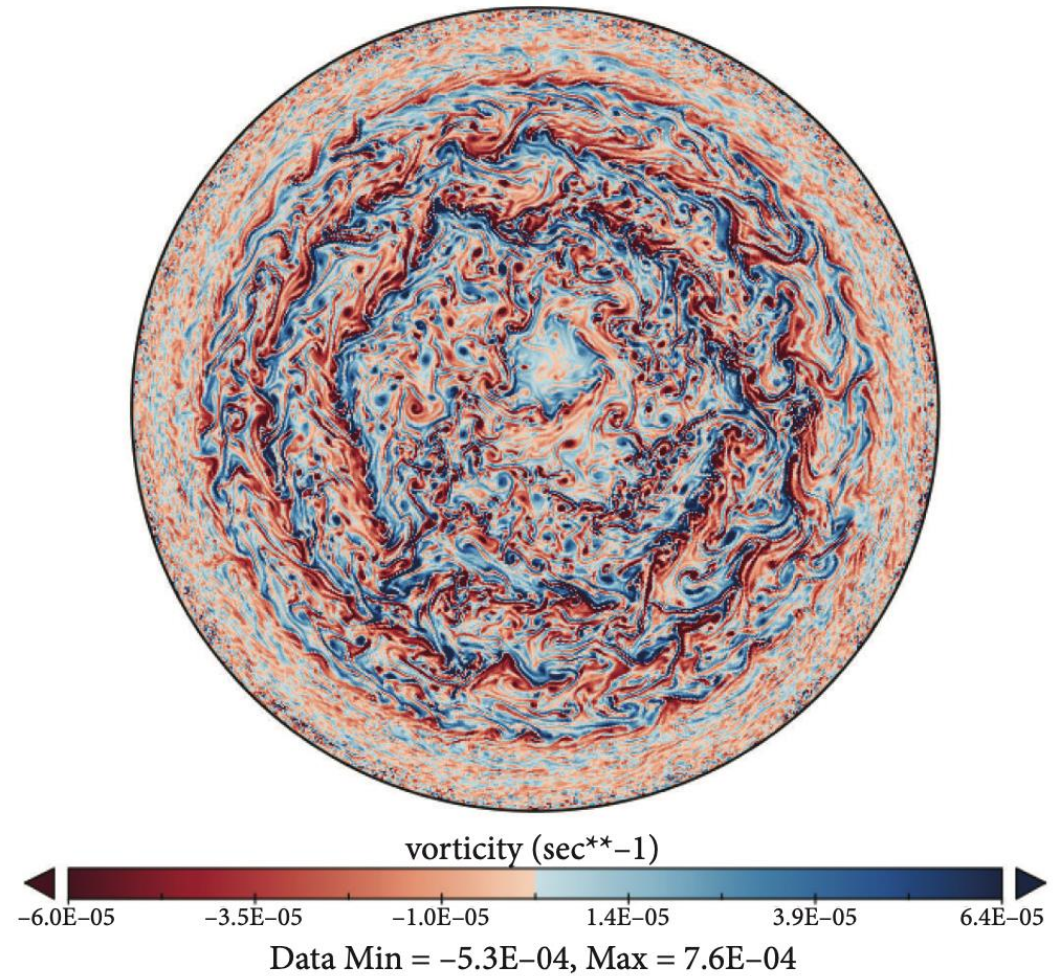
The mean flow is predictable. The perturbation is stochastic.



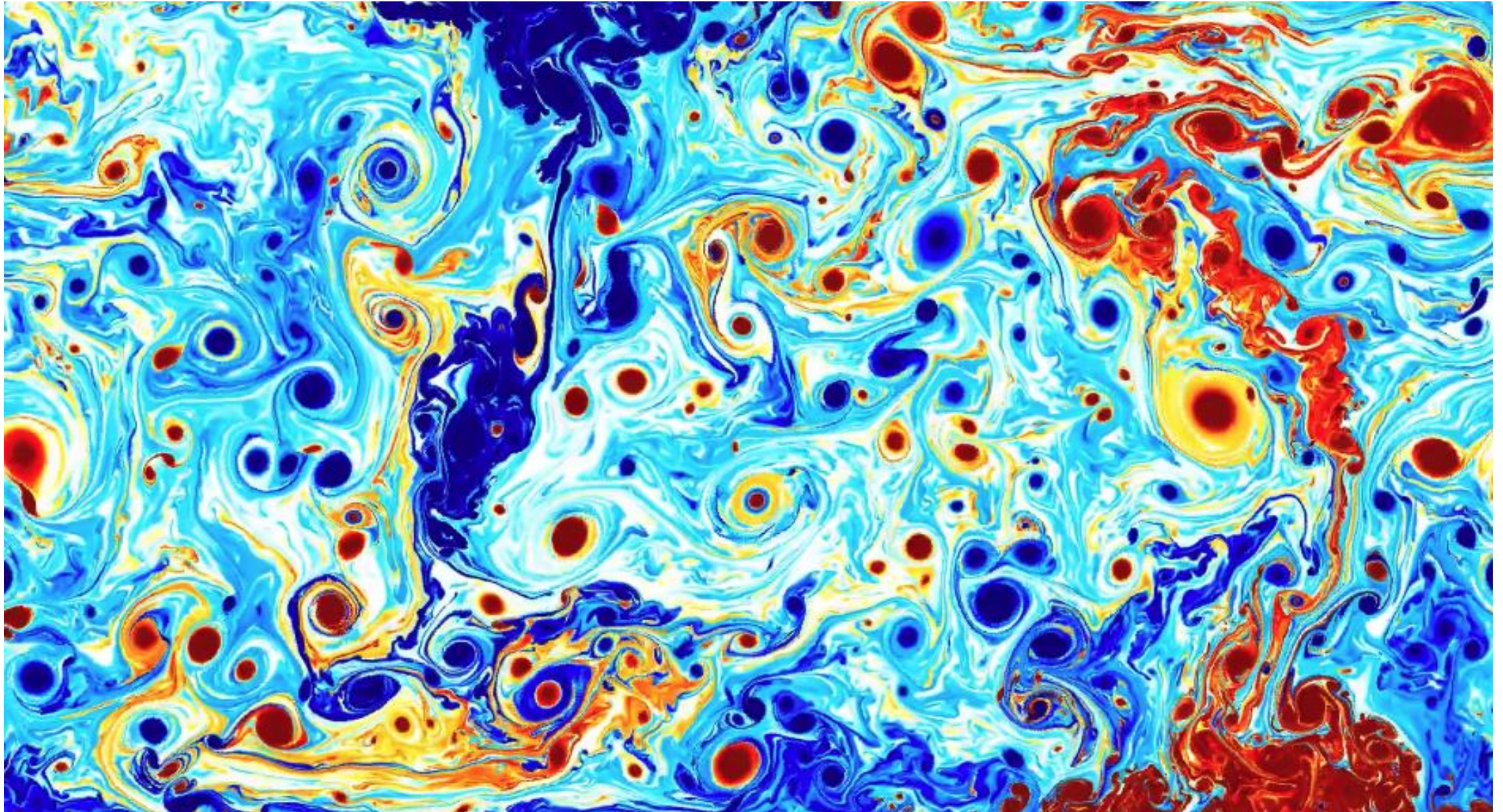
How to characterize the perturbations?

Simulation of the polar atmosphere of Jupiter.

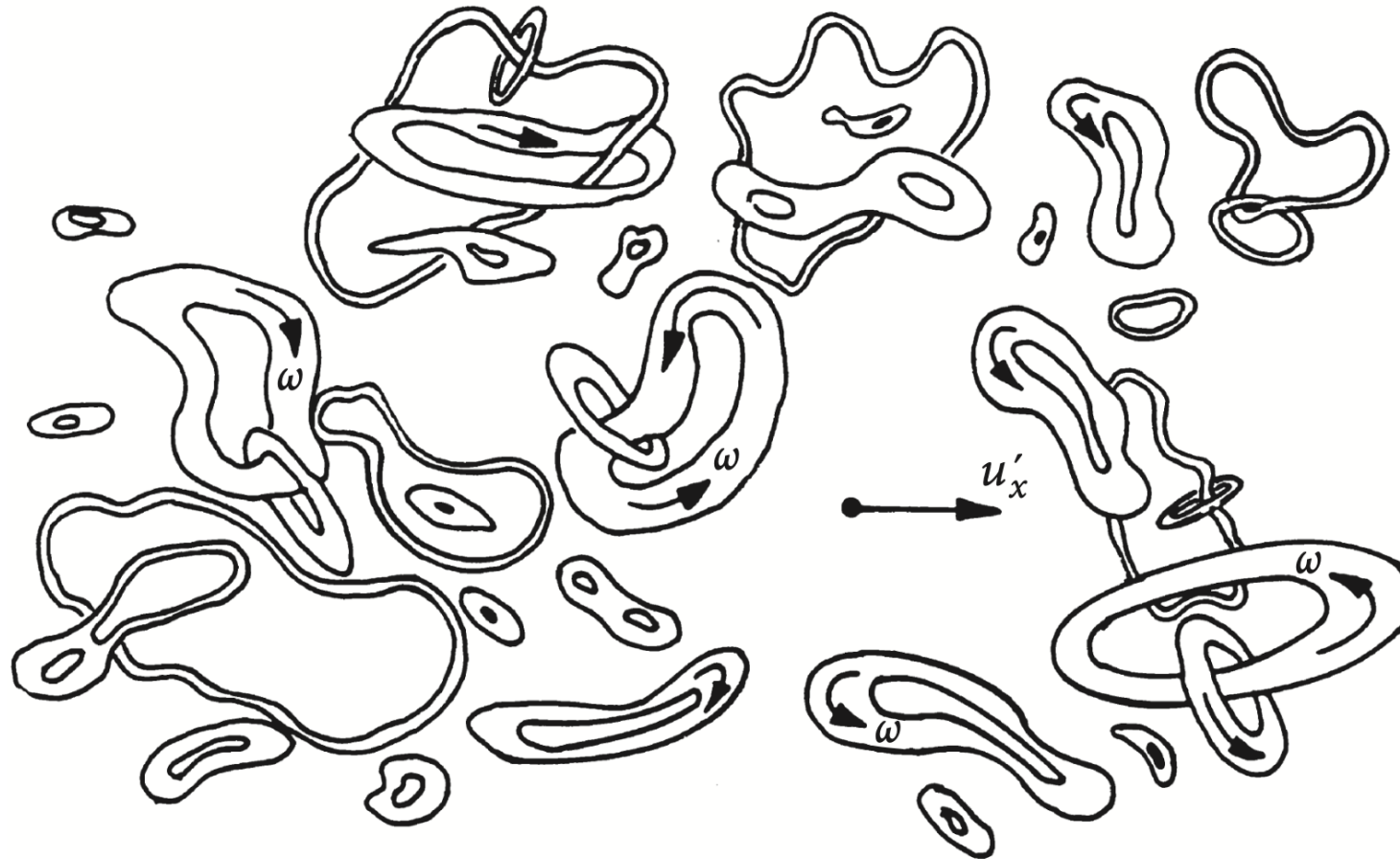
The flow consists of vortices of several different lengths.



Fully developed turbulence
Vortices are the main unit of turbulent flow.



Definition?

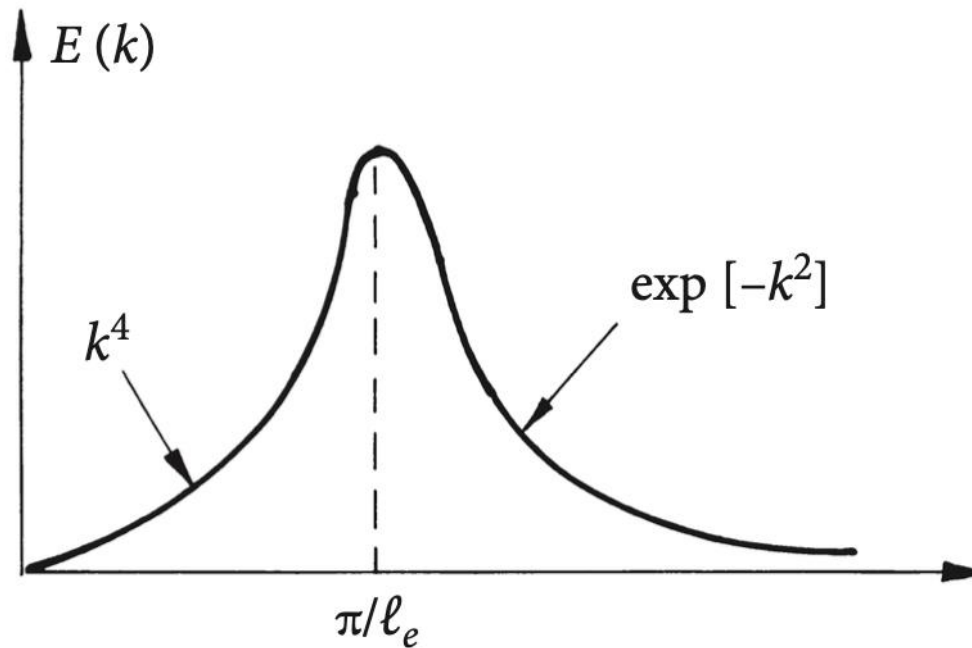


Turbulence: a tangle of interacting vortex tubes

Velocity and Vorticity

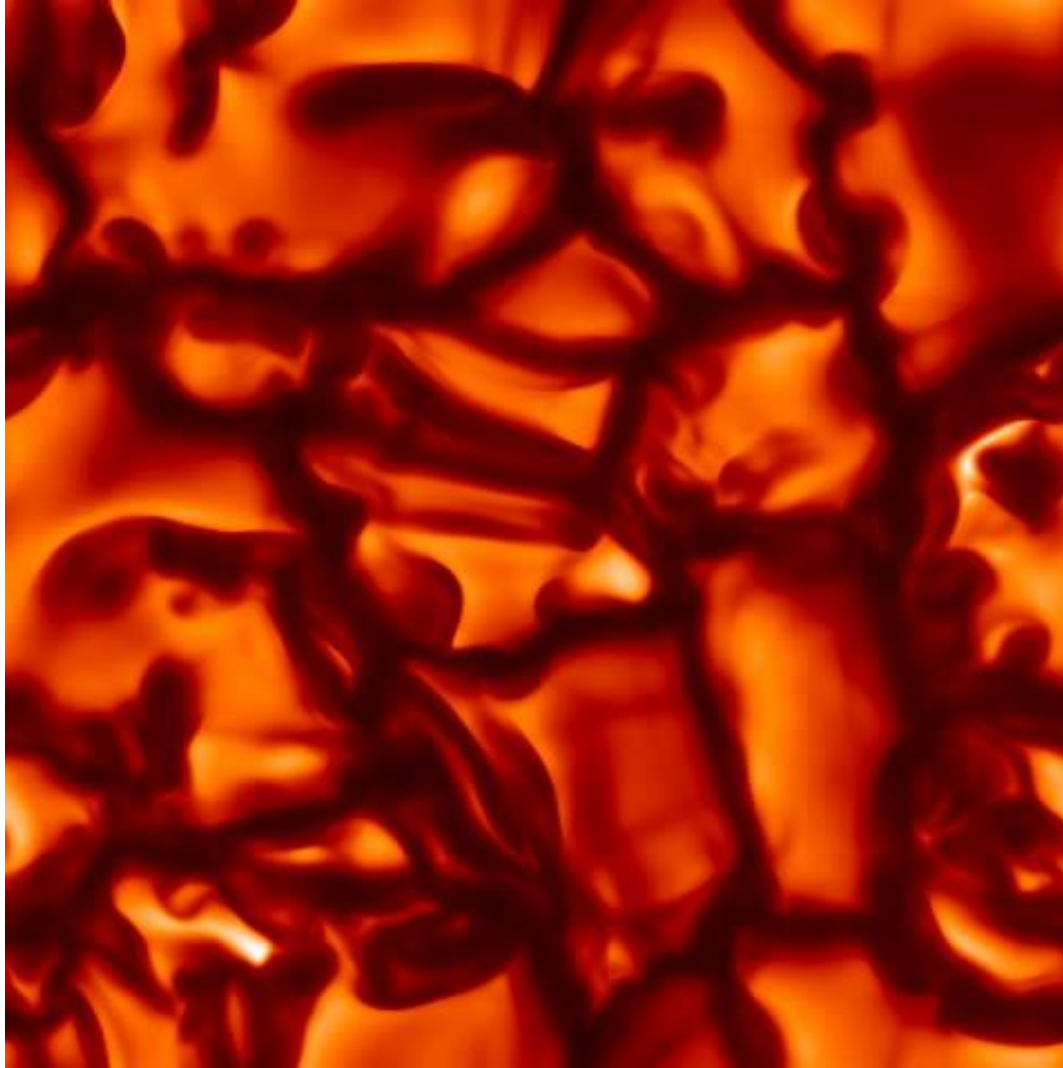
$$\mathbf{u}(\mathbf{r}) = \frac{1}{4\pi} \int \frac{\boldsymbol{\omega}(\mathbf{r}') \times \mathbf{r}}{r^2} d\mathbf{r}'$$

A vortex sets up a far-reaching velocity field.
This velocity field affects other vortices.



A vortex of size ℓ_e has its kinetic energy peaking around the same scales, but also tails that extend to larger and smaller wavenumbers.

Example: Turbulence in stars



Simulation of stellar
convection.

Surface intensity

$$T_{\text{eff}} = 4400 \text{ K}$$

$$L = 4.4 \times 10^6 \text{ km}$$

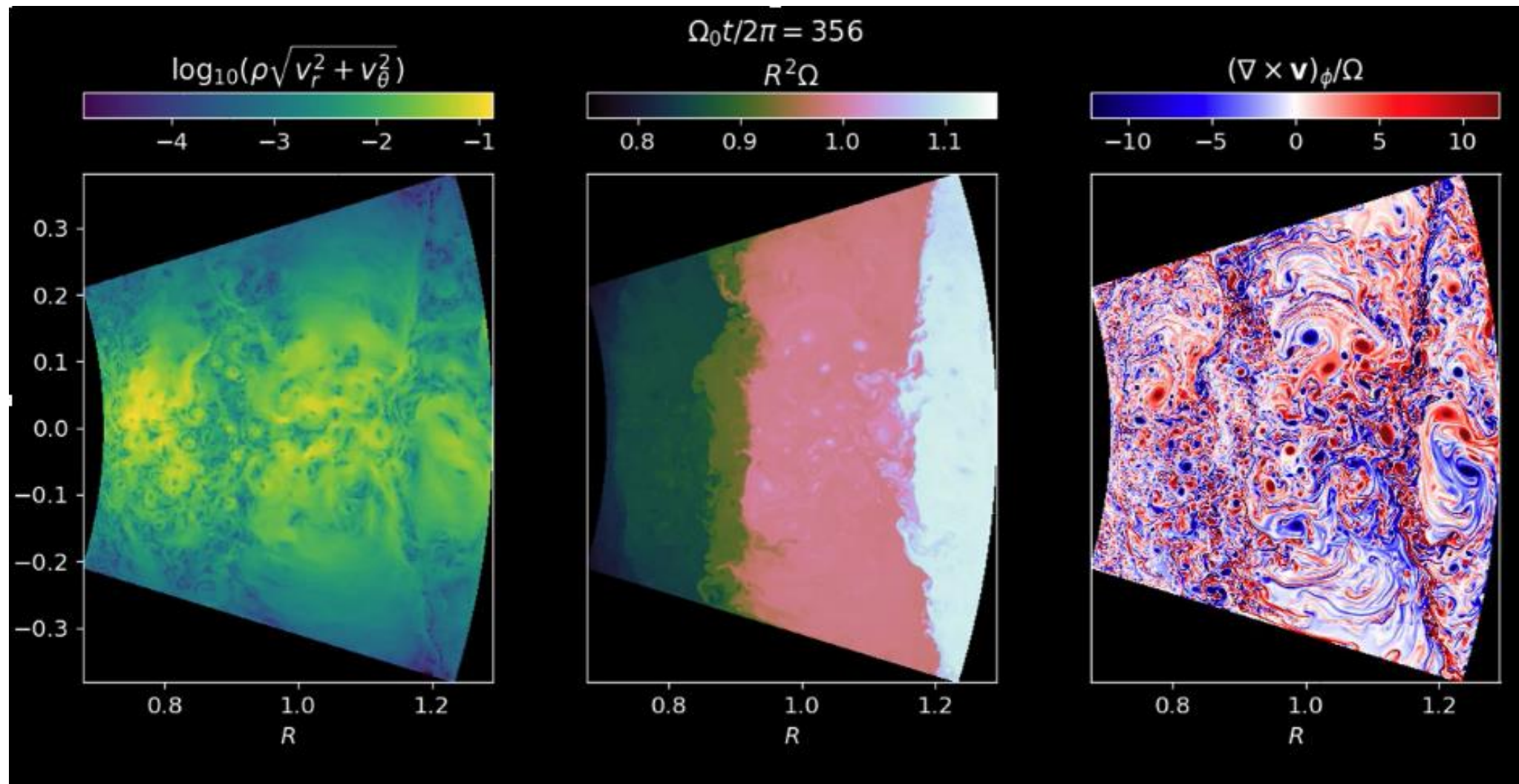
Example: Turbulence in stars

Same simulation, meridional plane cut.

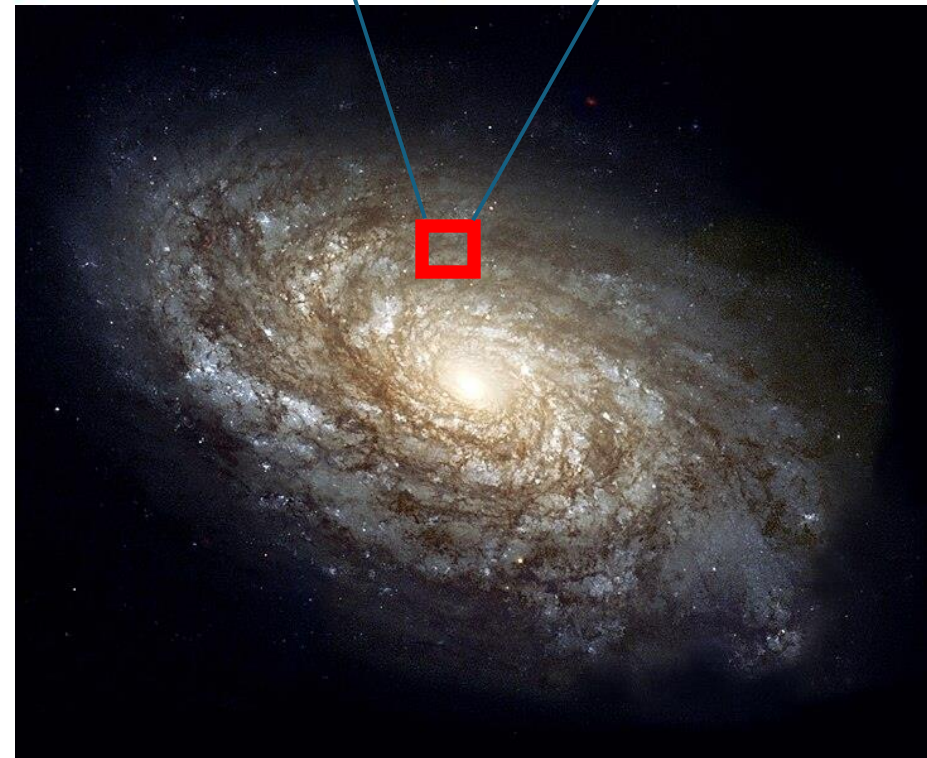
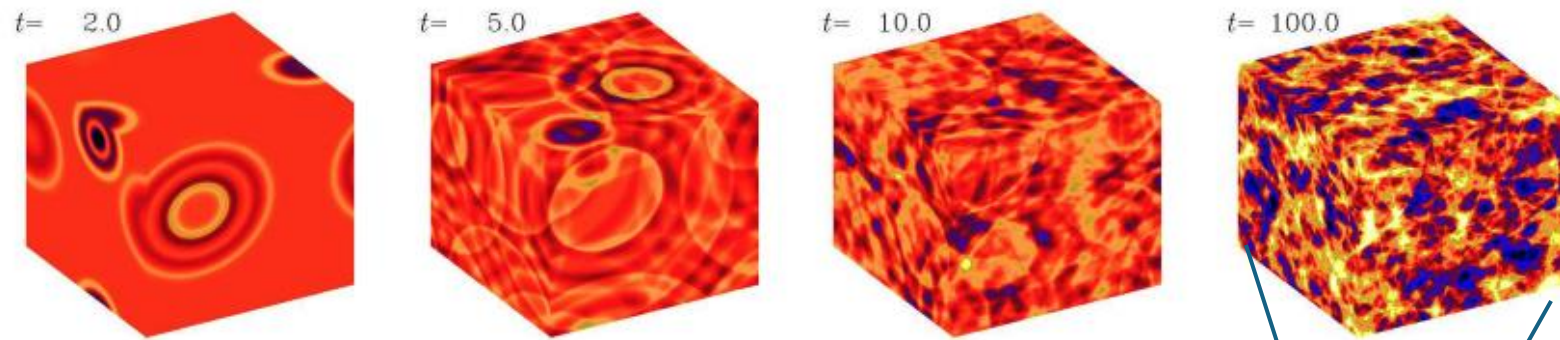


$$L_z = 2 \times 10^6 \text{ km}$$

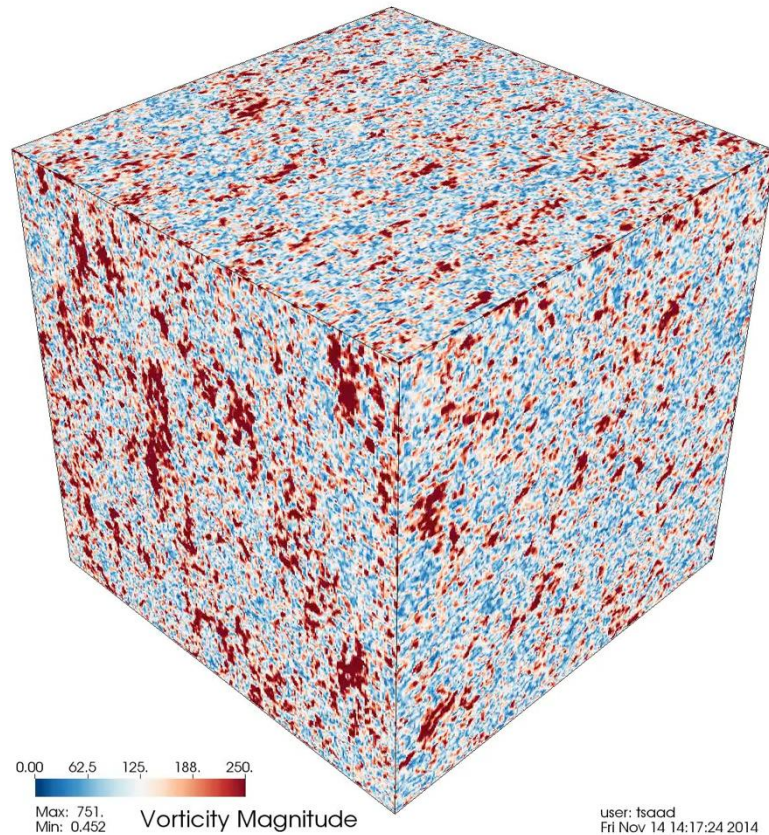
Example: Turbulence in accretion disks



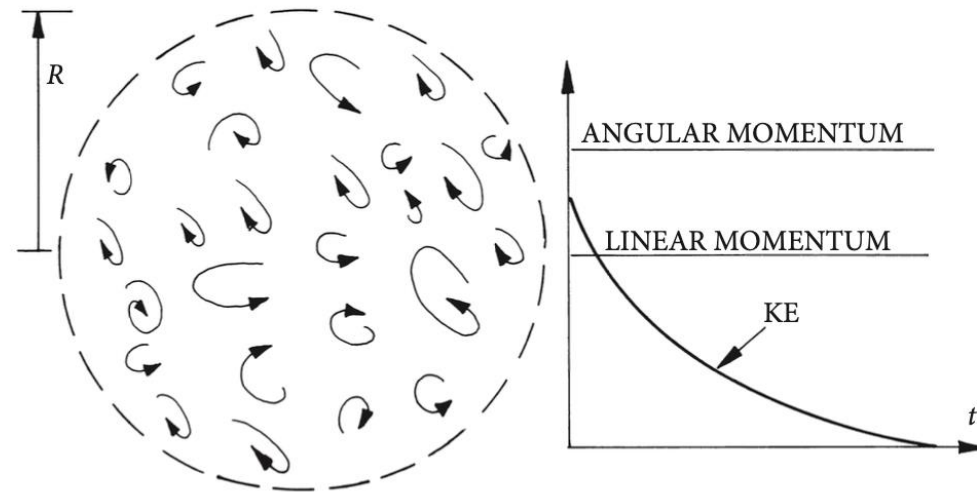
Example: Turbulence in the ISM



Freely decaying turbulence

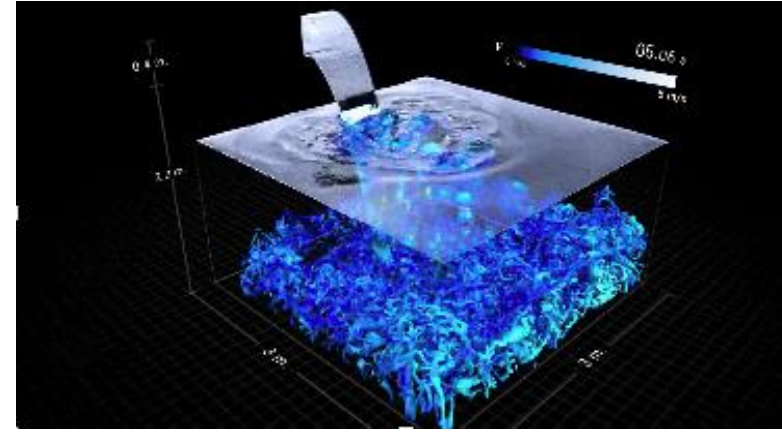


user: tsaad
Fri Nov 14 14:17:24 2014



- Constant angular momentum
- Constant linear momentum
- Decaying kinetic energy
- Rising temperature (if adiabatic)

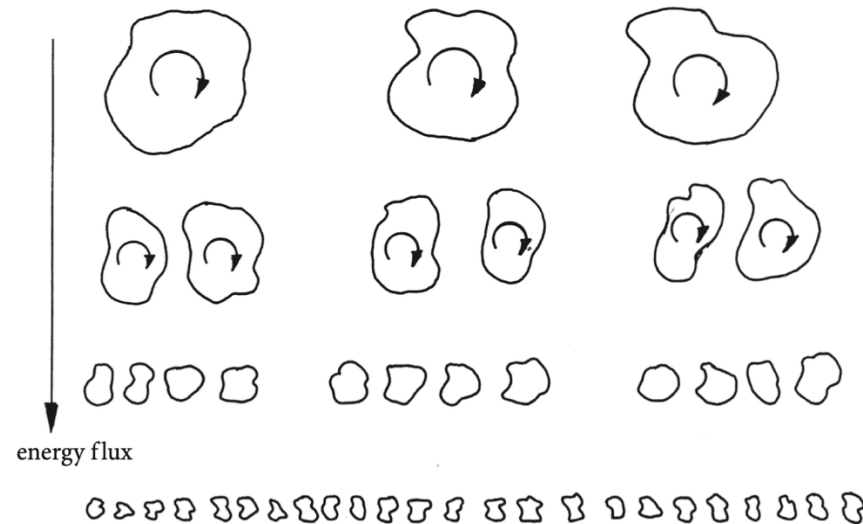
Energy injection



Cascade

Energy is injected at the large scales of the flow (in this case, the kinetic energy of the water hitting the surface).

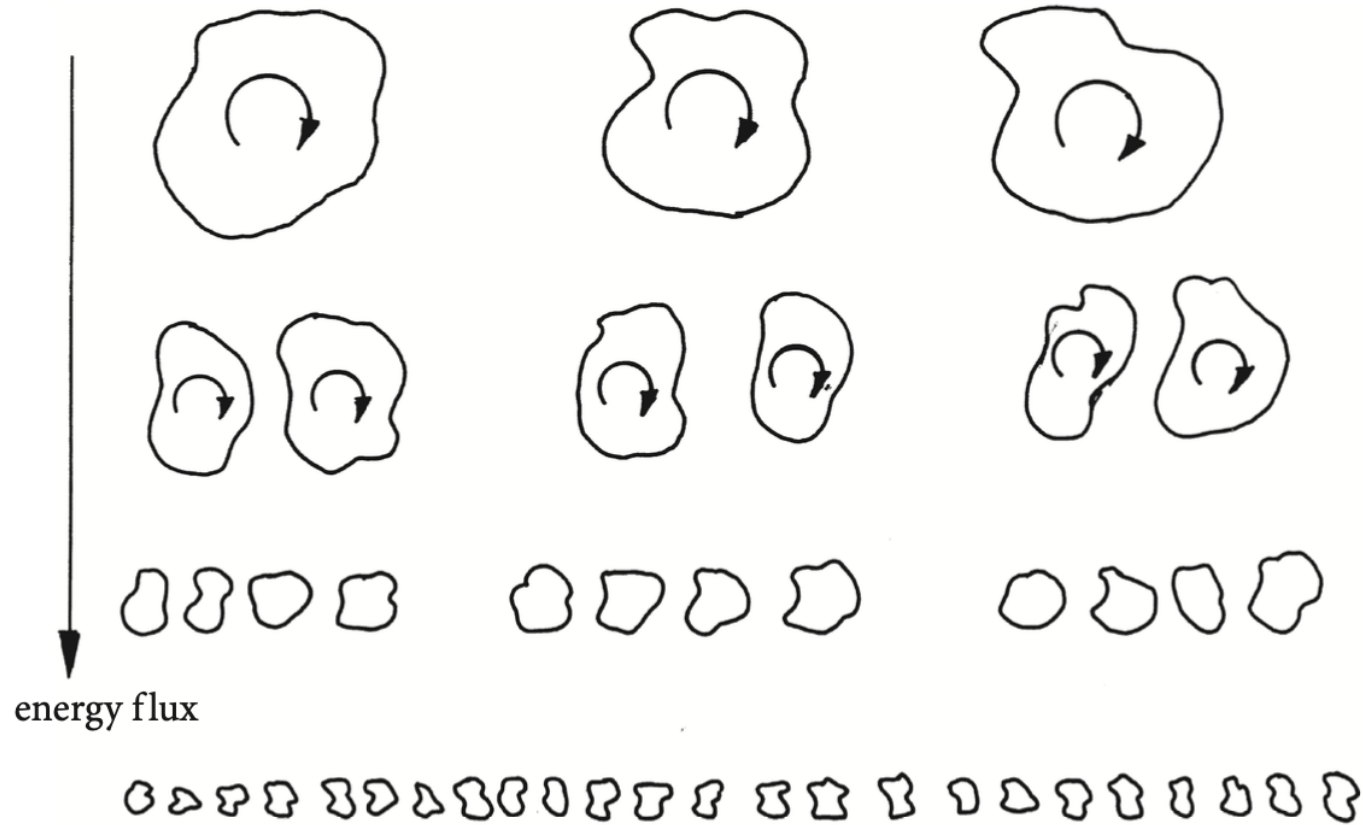
It then cascades toward smaller scales as eddies break down after a few turnover times.



Energy is injected at the large scales of the flow (in this case, the kinetic energy of the water hitting the surface).

It then cascades toward smaller scales as eddies break down after a few turnover times.

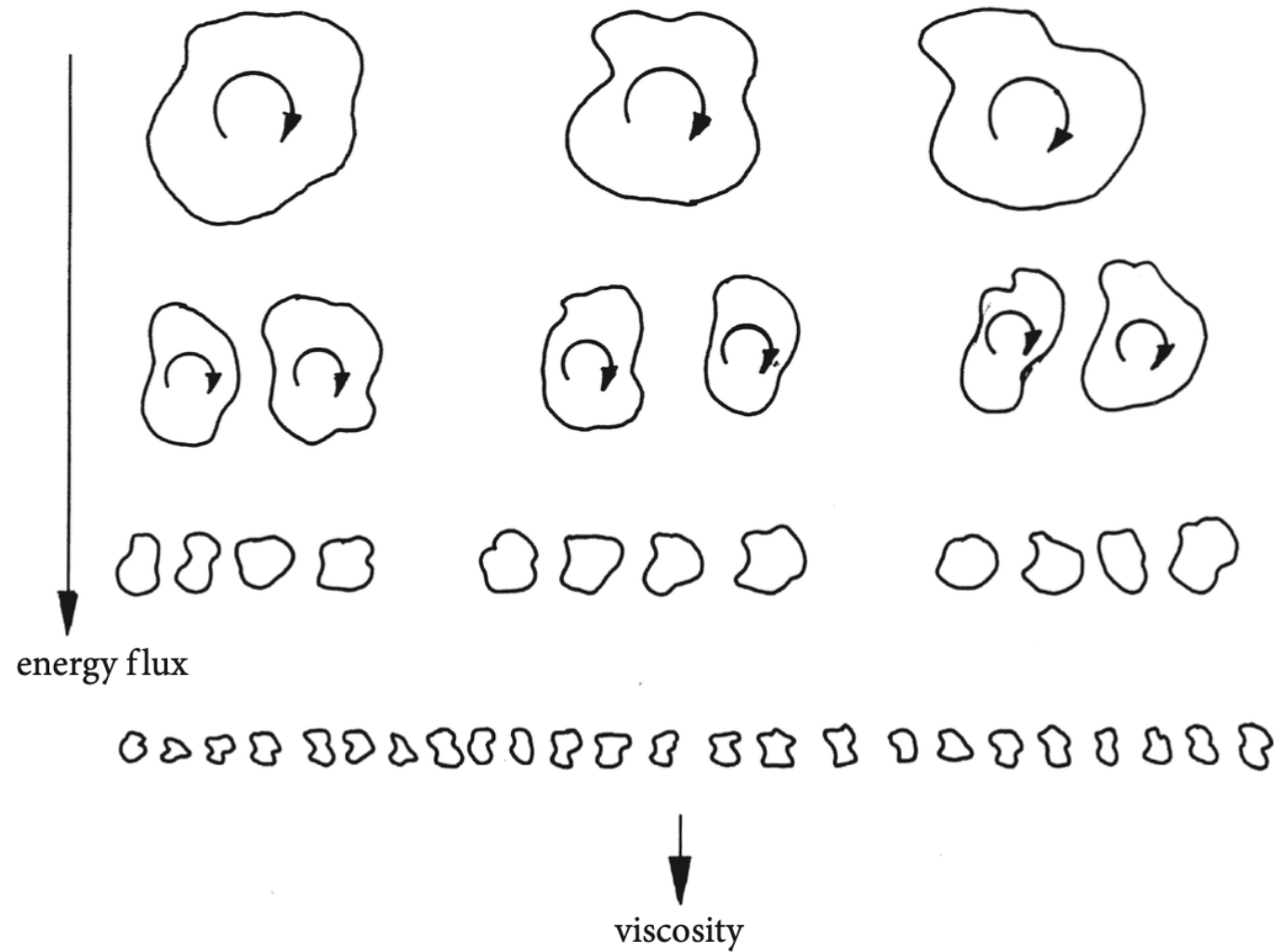
The turbulent cascade



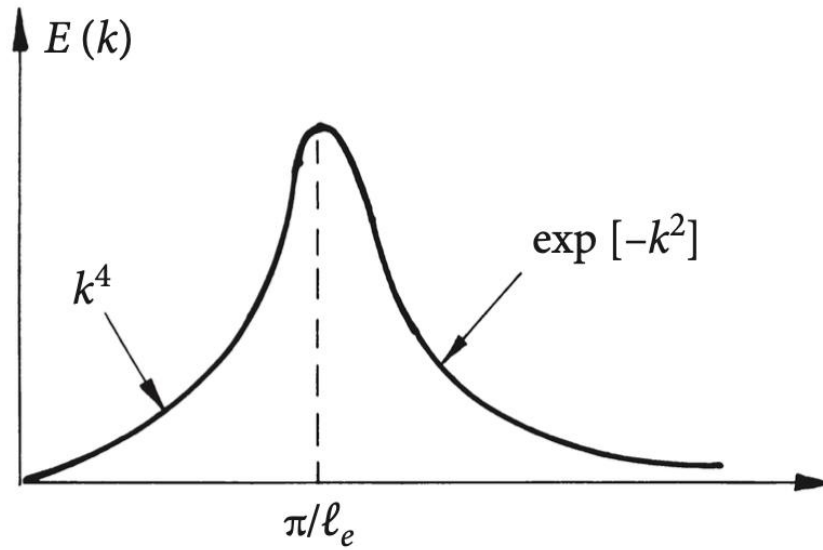
The cascade end is viscosity

*Big whirls have little whirls,
which feed on their velocity;
And little whirls have lesser
whirls,
And so on to viscosity.*

Richardson, 1922

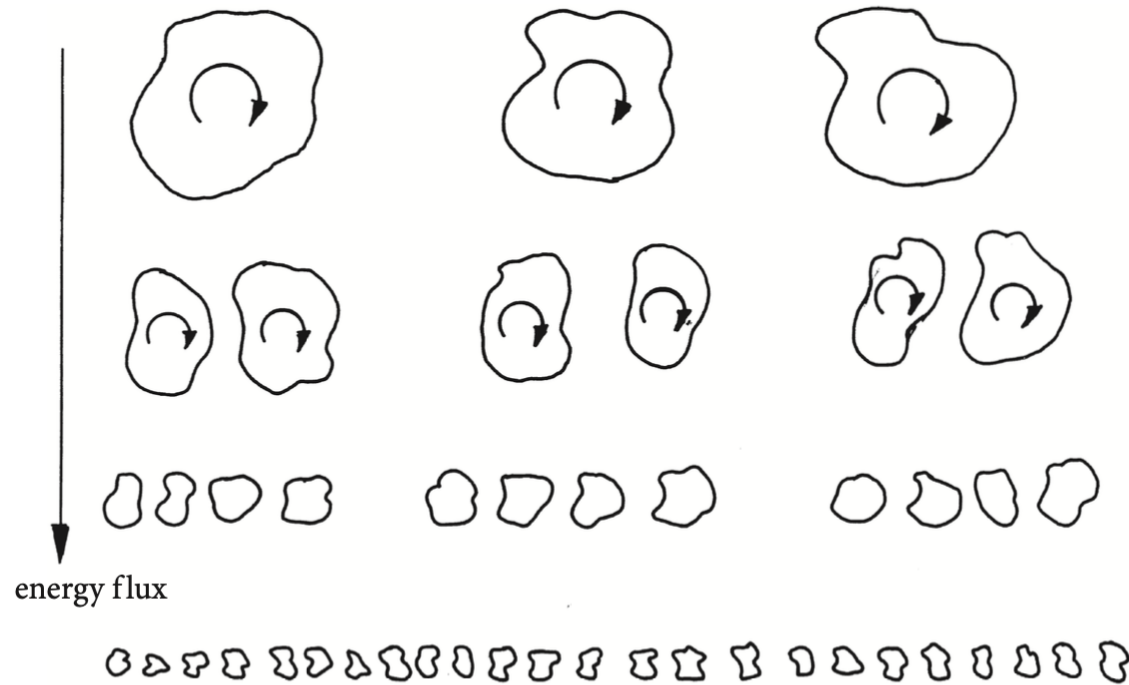


The cascade is local

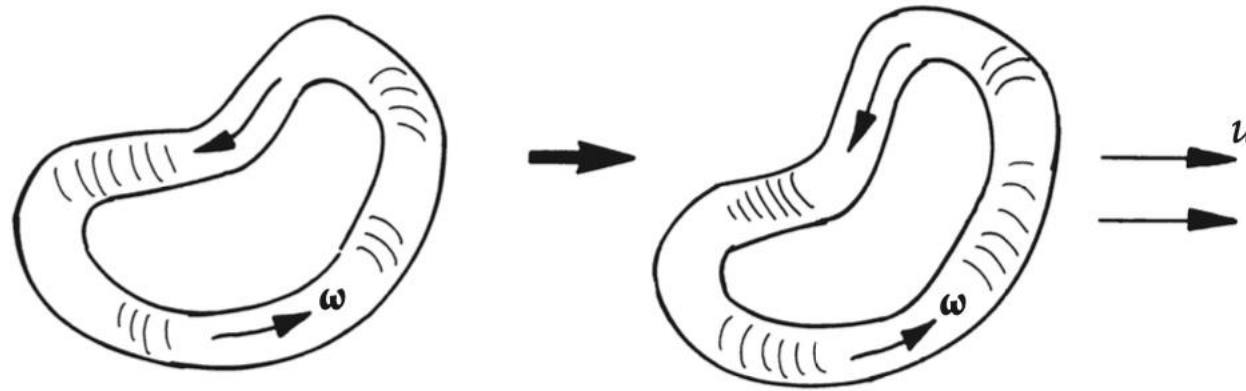


Eddies only distort other eddies of similar length.

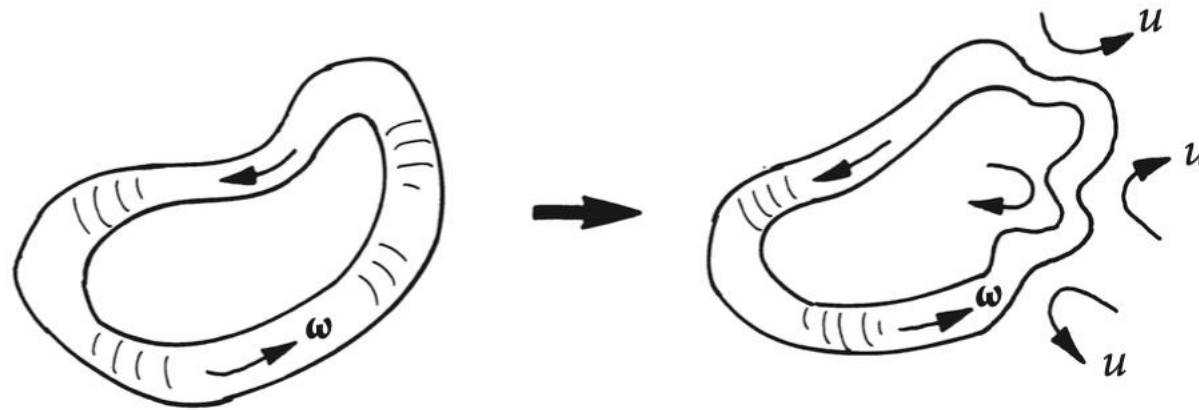
Why?



The cascade is local

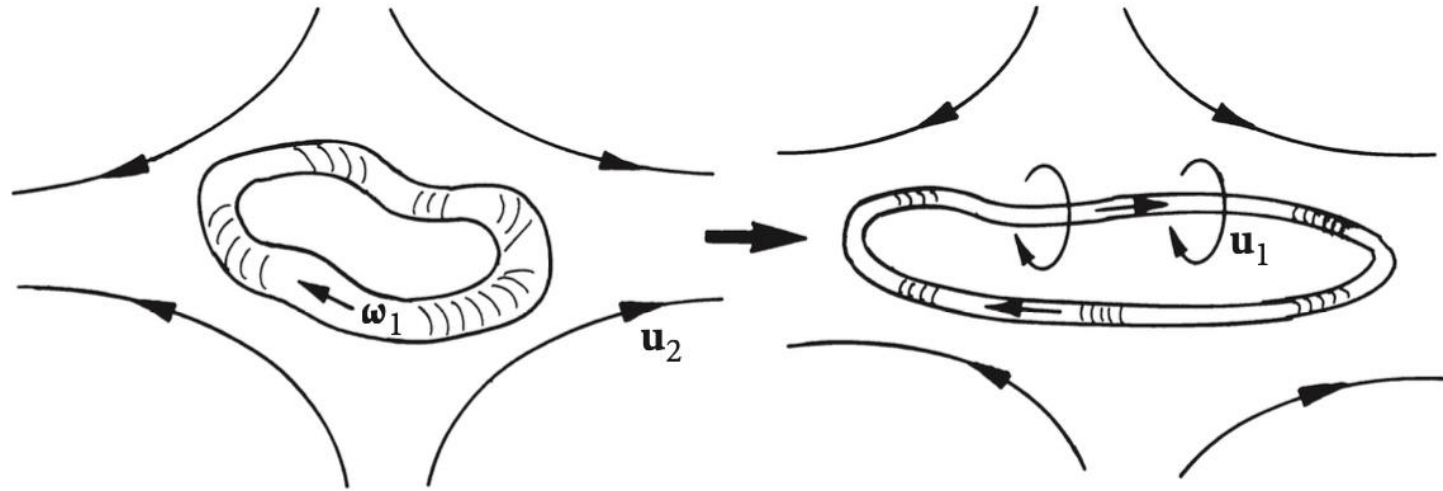


(a) large-scale velocity simply moves the vortex tube around



(b) small-scale velocity wrinkles surface of the vortex tube

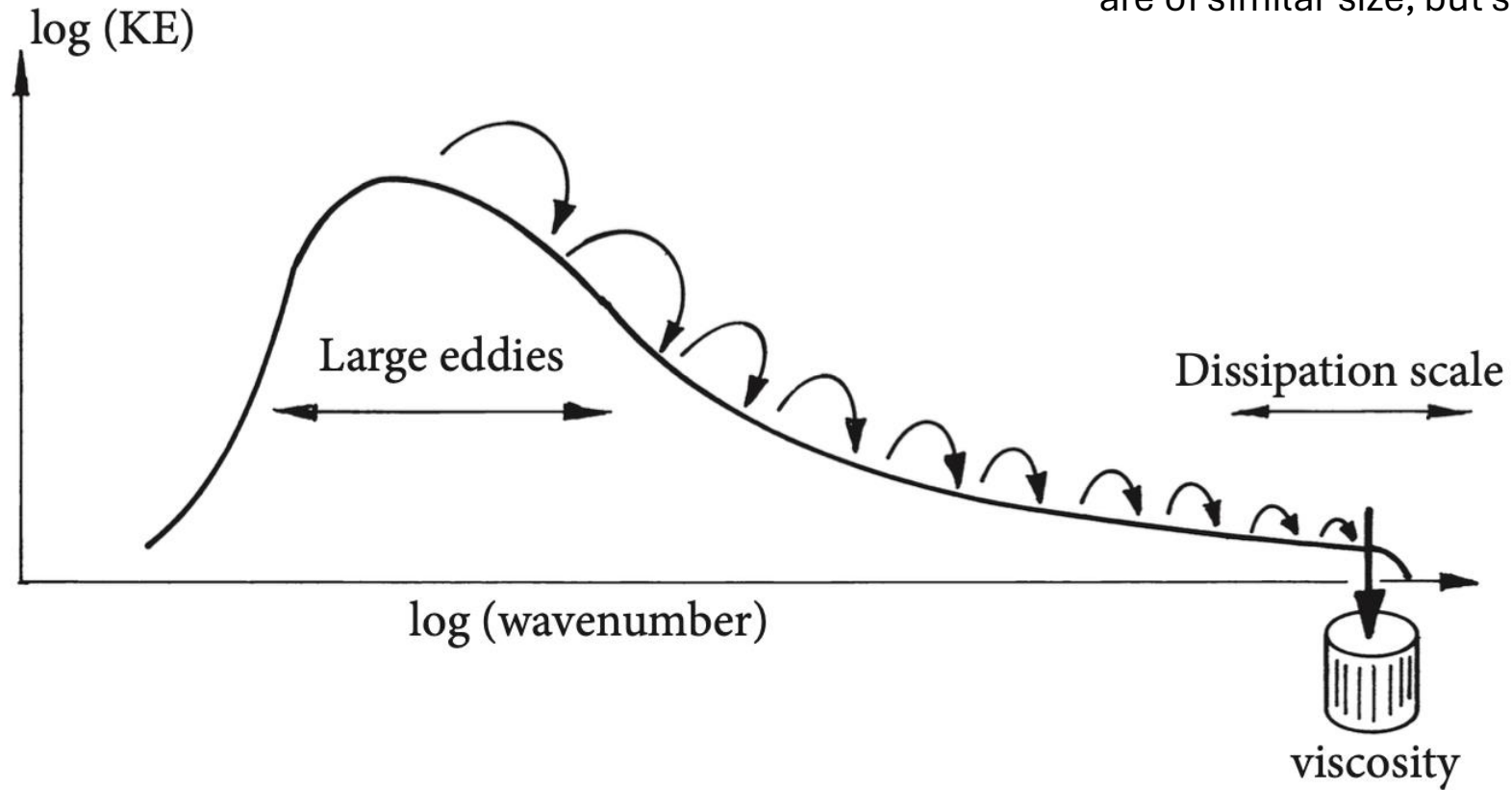
Stretching of vortex tubes



But in the strain field of vortices of similar size,
vortex lines get stretched.

The cascade is local and inertial

Via vortex stretching, a vortex transfers its energy to vortices that are of similar size, but smaller.



Viscosity is rather passive in the cascade,
It's like a dust bin, only mopping up energy when it reaches the viscous scale.
The cascade is *inertial*.

The Energy Spectrum: Kolmogorov's $k^{-5/3}$ law

Energy injection Π at the integral scale

Energy dissipation ε at small scale.

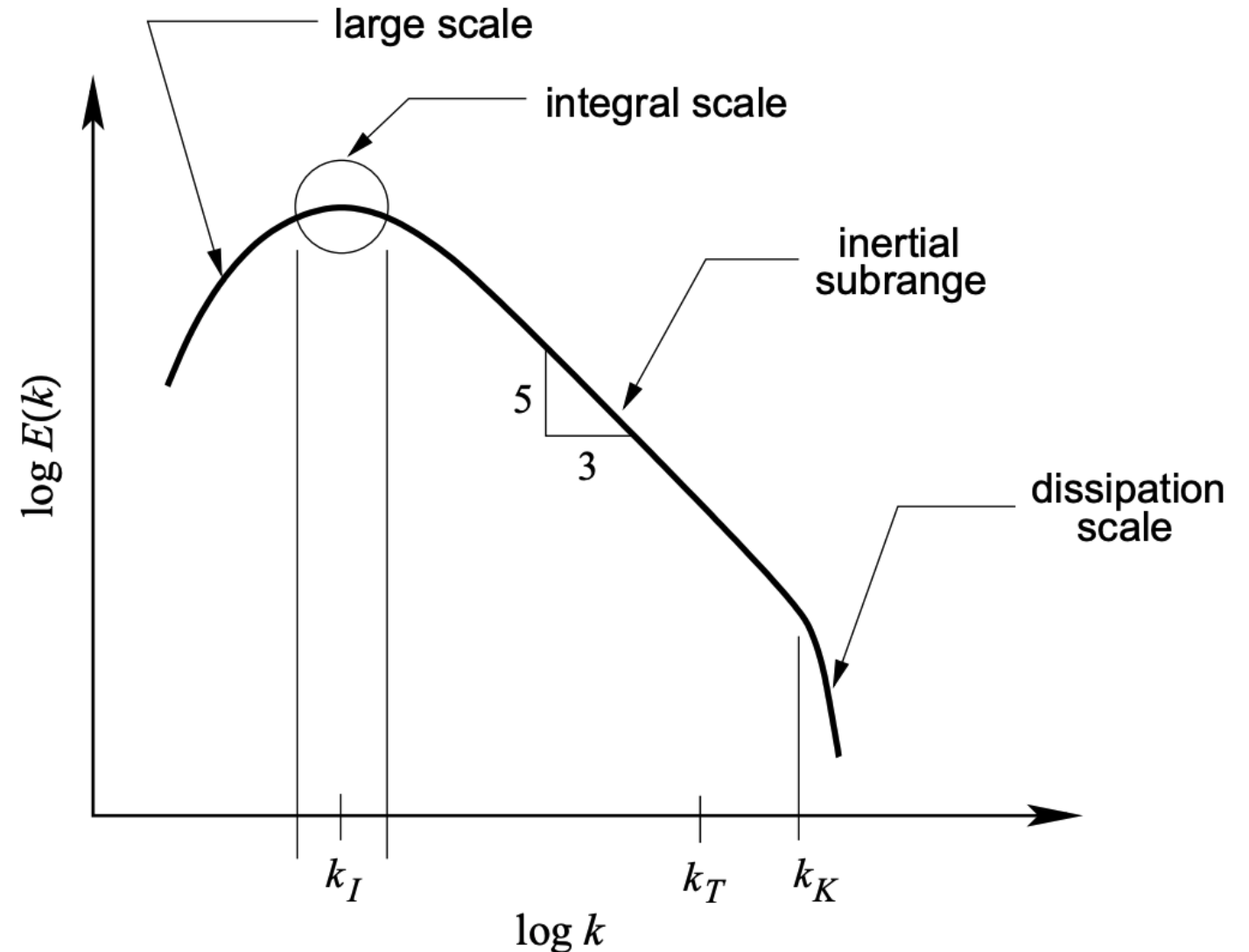
In steady state, $\Pi = \varepsilon$

The energy goes as $E(k)$ propto $k^{-5/3}$

Proof via dimensional analysis:

The variables that can shape $E(k)$ are only k and ε .

The only way k and ε can combine to give
The dimension of $E(k)$ (energy per
wavenumber) is via $E(k) = C \varepsilon^{2/3} k^{-5/3}$



Simulations of turbulence recover the $k^{-5/3}$ law

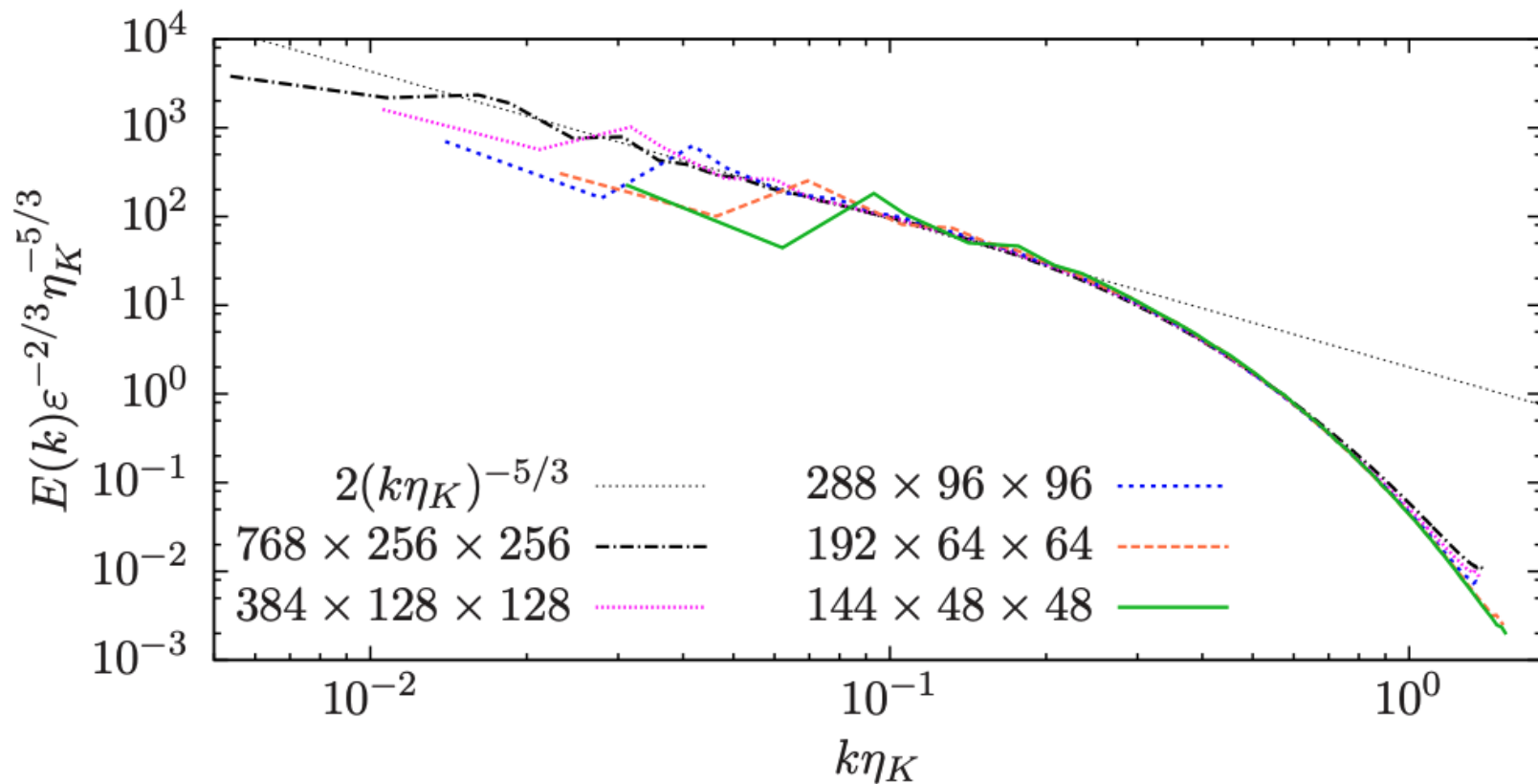
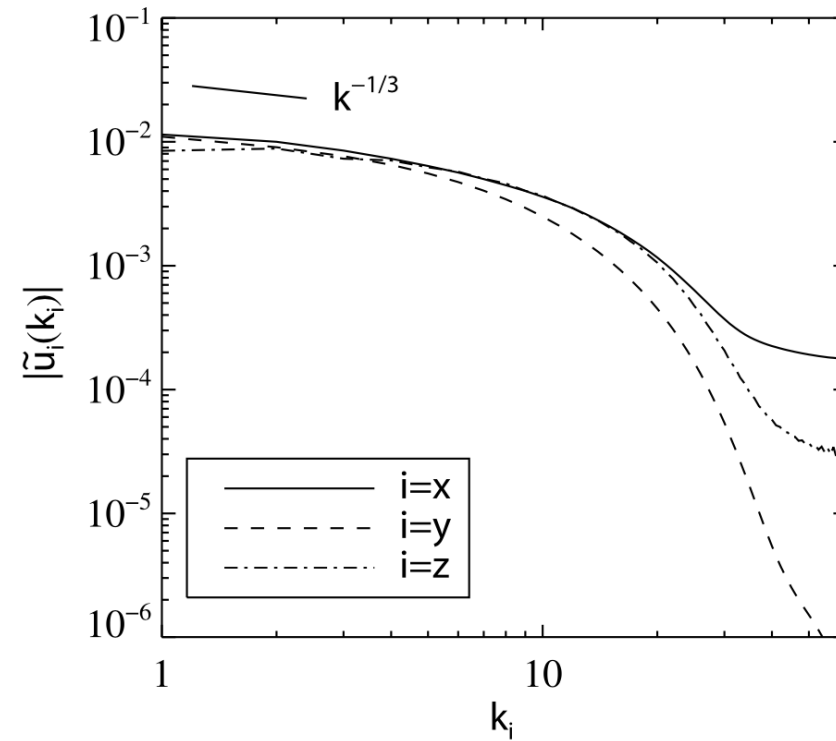
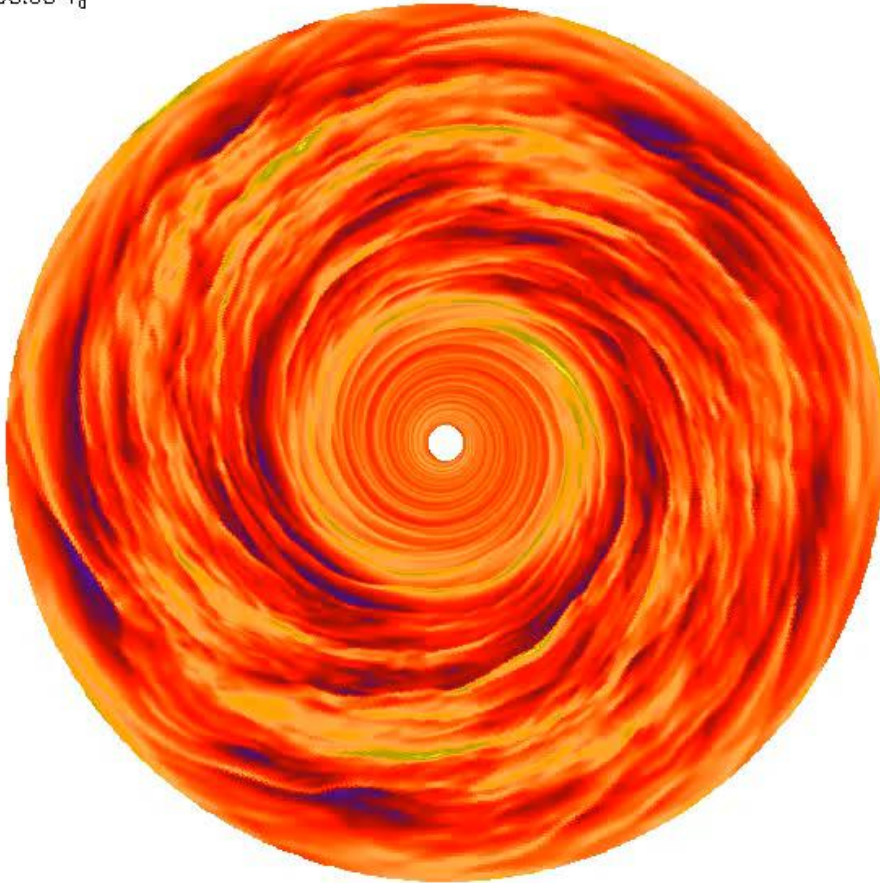


FIG. 1. Energy spectra of the Kolmogorov flow simulations for $n = 48, 64, 96, 128, 256$. The spectra are time-averaged over 6, 4.5, 3, 2.75, 1.5 large-eddy turnover times, respectively.

Turbulence in accretion disks show the $k^{-5/3}$ law

$t=95.58 T_d$



In terms of $u(k)$ instead of $E(k)$,
the scaling is $k^{1/3}$

Kolmogorov 1941 theory of small scales

Kolmogorov found how the small scale velocity and length u, l of the flow in terms of the large scales U, L

In steady state, $\Pi = \varepsilon$

$$\Pi \sim U^3/L$$

$$\varepsilon \sim \nu (u/l)^2$$

At the large scales, $Re = LU/\nu$

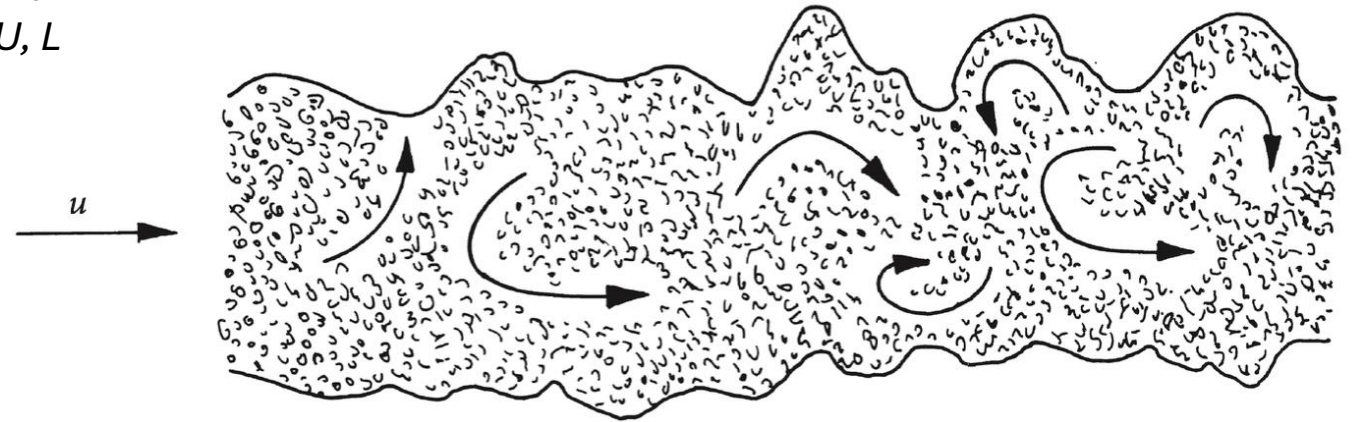
At the small scales, the Reynolds number is 1, so $\nu \sim lu$.

Combining gives

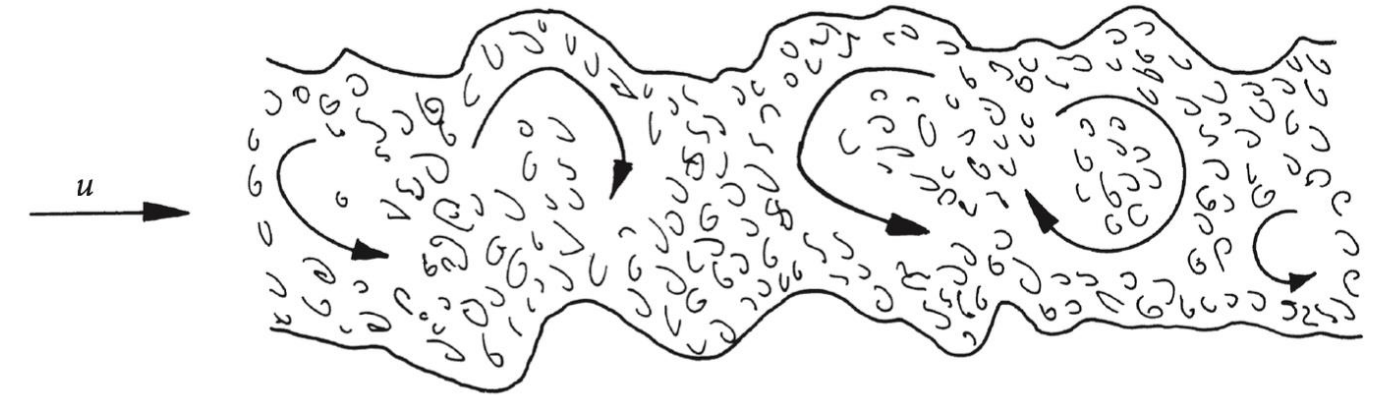
$$l \sim LRe^{-3/4}$$

$$u \sim URe^{-1/4}$$

As Re decreases, the dissipation range moves toward larger scales.



High Re



Modest Re

Reynolds averaging

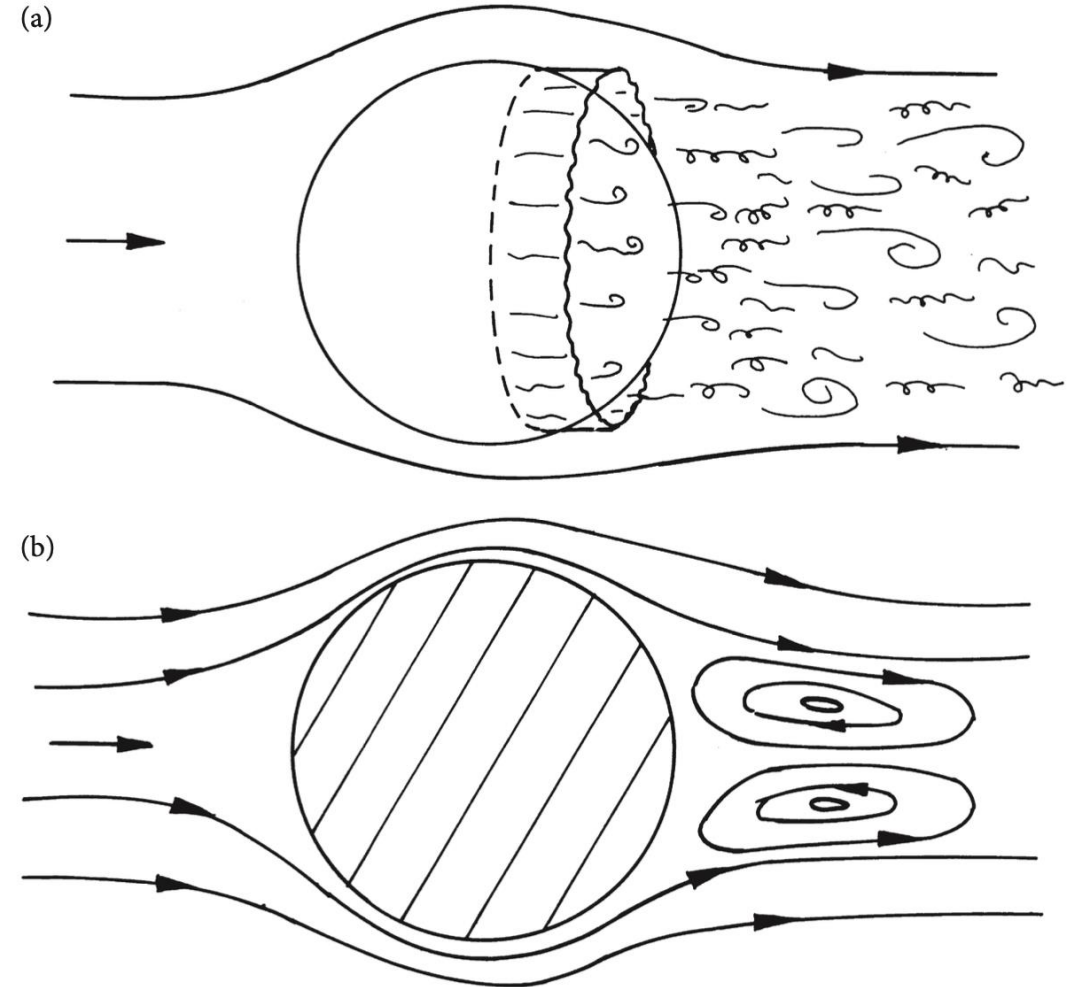
Decompose the flow into mean flow and perturbation
Average to find an equation for the mean flow, the
Reynolds-Averaged Navier-Stokes (RAMS):

$$\frac{\partial \rho \bar{u}_i}{\partial t} + \partial_j (\rho \bar{u}_i \bar{u}_j) = -\partial_i \bar{p} - \partial_j (\langle \rho \delta u_i \delta u_j \rangle)$$

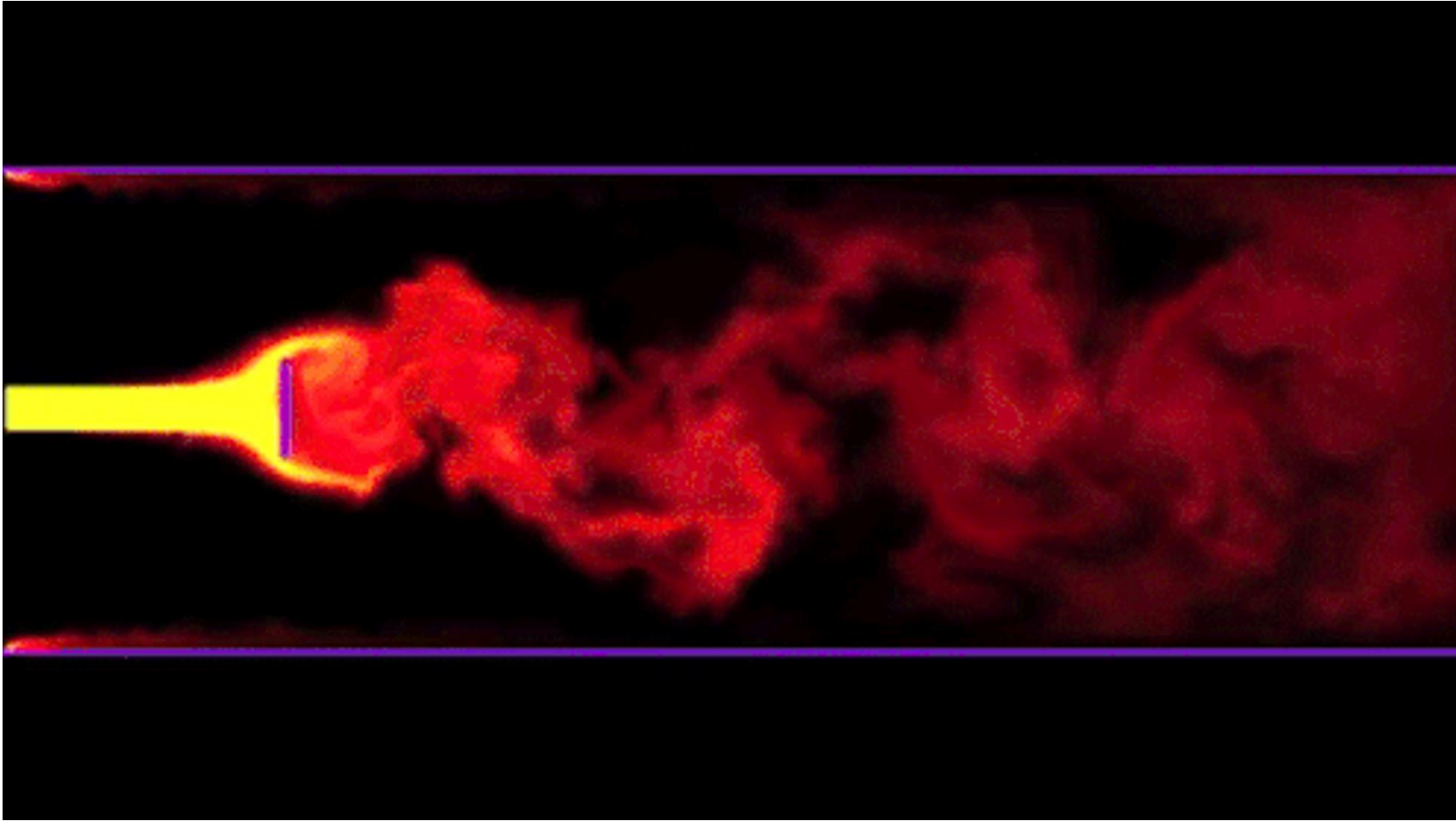
The extra term in the RHS is an extra ram pressure,
due to the small scale fluctuations.

This is the Reynolds tensor. Notice that it behaves
like a viscous term. The turbulence diffuses
momentum out of a control volume.

This is the ***turbulent viscosity***. It couples the mean
flow to the turbulence.



Turbulent Viscosity



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This is the Reynolds tensor. Notice that it behaves like a viscous term. The turbulence diffuses momentum out of a control volume.

This is the ***turbulent viscosity***. It couples the mean flow to the turbulence.

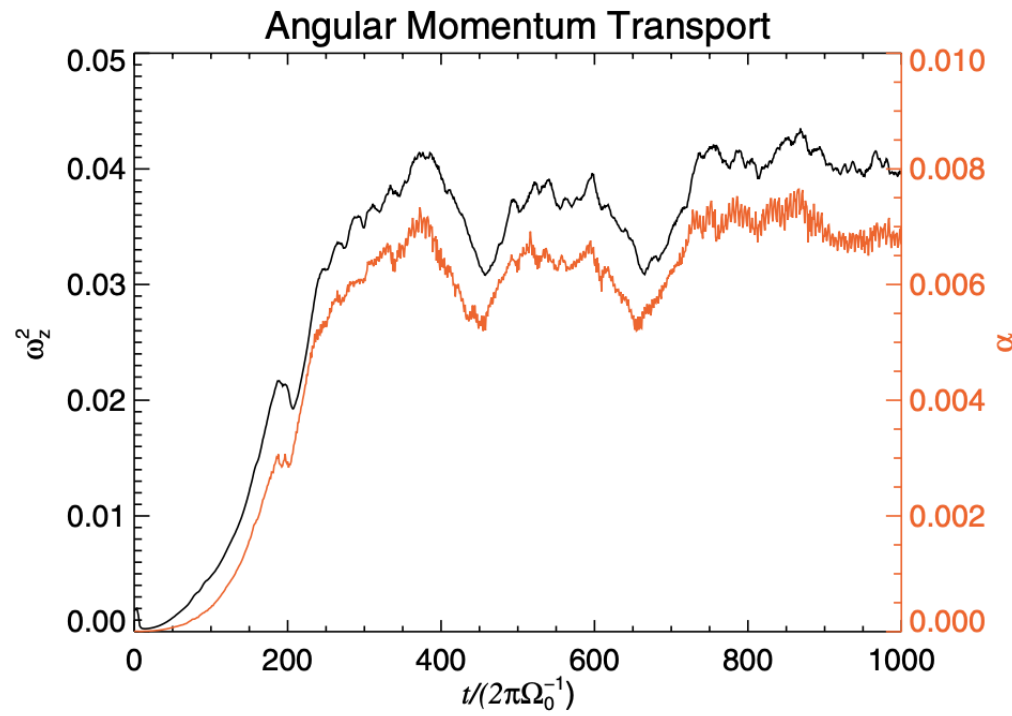


Fig. 3. Enstrophy and the resulting alpha-viscosity for the fiducial 2D run. The two quantities are quite well correlated, since the angular momentum transport is the result of inertial-acoustic waves, which in turn are driven by vorticity.

flow. We measured the kinetic alpha values in the simulations,

$$\alpha = \frac{R^{xy}}{qp_0} \quad (14)$$

where $R^{xy} = \overline{\rho\delta u_x\delta u_y}$ is the Reynolds stress. The value measured is $\alpha \approx 5 \times 10^{-3}$, indicating good transport of angular momentum. The temporal variation of alpha (Fig. 3) matches that of the enstrophy.